THE UNIVERSITY of EDINBURGH

Comparative Analysis of Wrist and Finger PPG Signals for Vascular Health and Heart Rate Monitoring

Participant Information Sheet

You are being invited to take part in a research study. Before you decide whether or not to take part, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully. Talk to others about the study if you wish. Contact us if there is anything that is not clear, or if you would like more information. Take time to decide whether or not you wish to take part.

What is the purpose of the study?

We wish to study the differences between heart rate signals recorded from the wrist and the fingertip. The aim is to determine which measurement site provides more accurate information for assessing vascular health and heart rate, especially when moving or breathing differently. There are no direct benefits to you taking part in this study. In the future, results from this study might help improve the accuracy of wearable health monitoring devices.

Why have I been invited to take part?

You are invited as a healthy volunteer to help us compare the accuracy of data from various sensor locations.

Do I have to take part?

No, it is up to you to decide to take part. If you decide to take part, you will be given this information sheet to keep and be asked to sign a consent form. If you decide to take part, you are still free to withdraw at any time and without giving a reason, and all your data will be destroyed.

What will happen if I take Part?

We will arrange a time for you to participate at the School of Engineering. The whole procedure should take approximately 15 minutes. You will be asked to wear two optical sensors: one on your fingertip and one on your wrist. We will record signals from these sensors under two short conditions: while you are resting comfortably and while performing paced breathing exercises.

Will my taking part be kept confidential?

All the information we collect during the research will be kept confidential and we follow strict laws which safeguard your privacy at every stage. Your data will be anonymized for analysis. This data will only be used for my Bng thesis and will be deleted after submission in April.

Researcher Contact Details

If you have any further questions about the study please contact Mehai Singh Charan at s2427836@ed.ac.uk

Please keep this sheet. Thank you for reading it.

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Research Consent Form

Name of Researcher: Mehai Singh Charan – s2427836@ed.ac.uk

Further information is available from: Dr. Stewart Smith Stewart.Smith@ed.ac.uk

List the procedures to be used in the study:

Optical wearable sensors (photoplethysmography/PPG) will be placed on the wrist and fingertip to record heart rate signals. Participants will be asked to undertake two test conditions: sitting at rest and performing paced breathing. The total duration of the procedure is approximately 15 minutes.

- I agree to participate in this study
- I have read this consent form and Information Sheet and had the opportunity to ask questions about them
- I understand that I am under no obligation to take part in this study
- I understand that I have the right to withdraw from this study at any stage, without having to provide a reason
- I understand that this is non-therapeutic research from which I cannot expect to derive any benefit
- I understand that the data will be curated subject to General Data Protection Regulation (GDPR) regulations and the Data Protection Act of 2018, and I am aware that the investigators are obliged to report any evidence of abuse or criminal activity (if this is found)

Name of research participant: .....

Signature of participant: .....

Signature of Investigator: .....

Date: .....

Two copies of each the Information sheet and Consent form to be made :
- One copy to be retained by Investigator and Second copy by the subject